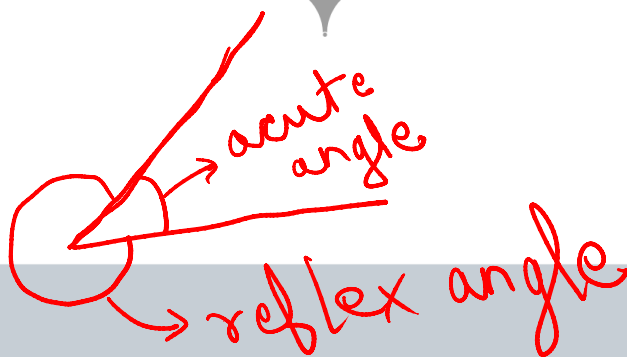
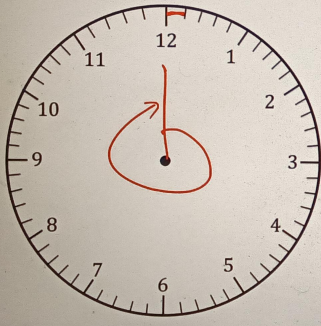


Clocks

CONCEPT



	MINUTE HAND	HOUR HAND
ONE HOUR	360°	30°
ONE MINUTE	6°	0.5°

$$\frac{360^\circ}{60} = 6^\circ$$

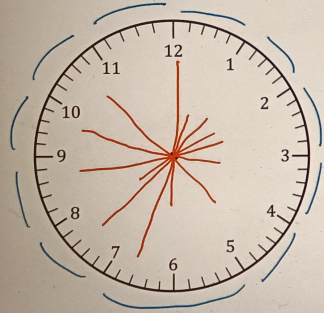
$$\frac{360}{12} = 30^\circ$$

$$\frac{30^\circ}{60} = 0.5^\circ$$

Dist. travelled by a minute hand in 1 hr = 360° & so on.

1. How many times in a day, are the hands of the clock in the same line, opposite direction?

- A. 22 B. 20 C. 24 D. 21

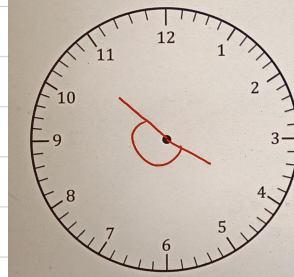


12 hrs → 11 times
24 hrs → 22 times

(coincide wale me bhi same ans hoga)

2. An accurate clock shows 4 o'clock in the evening. Through how many degrees will the hour hand rotate when the clock shows 10:45 p.m. on the same day?

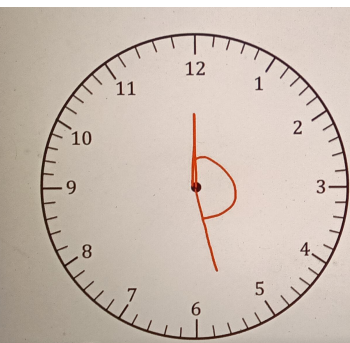
- A. 200.50° B. 202.45° C. 202.50° D. 202.63°



4:00 — 10:45
6 hrs 45 min
 $6 \times 30^\circ + 45 \times 0.5^\circ = 180 + 22.5 = 202.5$
(makes it ez to remember)

3. The wall clock in my home was stopped yesterday morning and restarted at noon. What is the angle swept by the hour hand when it was 20 minutes for 6?

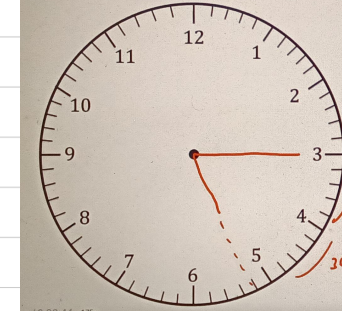
- A. 150° B. 160° C. 170° D. 180°



12:00 — 5:40
5 hrs 40 min
 $5 \times 30^\circ + 40 \times 0.5^\circ = 150^\circ + 20^\circ = 170^\circ$
or just remember the hour etc.
(40 x 30 / 60)

4. At what angle the hands of a clock are inclined at 5:15 in the morning on Saturday?

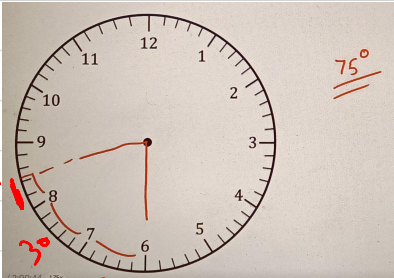
- A. 60° B. 65° C. 67.5° D. 72.5°



15 min x 0.5° = 7.5
67.5

5. What was the angle between the needles of the clock at 8:30 a.m. this morning?

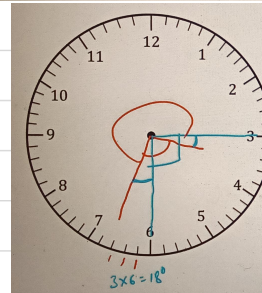
- A. 60° B. 74.5° C. 75° D. 75.5°



15°
30°
75°

6. What is the reflex angle between the short and the long hands of the clocks at 3:33 a.m. on 13th September?

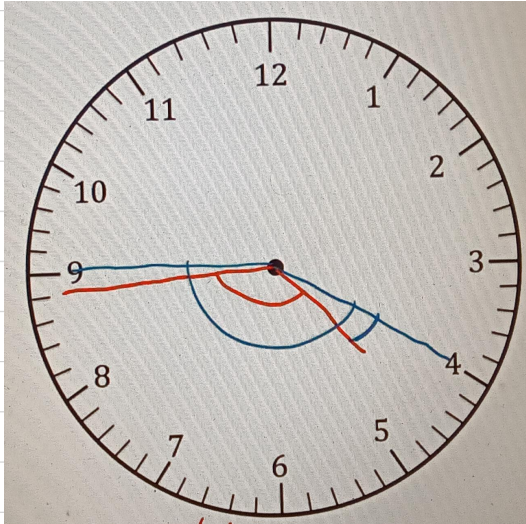
- A. 91.5° B. 268.5° C. 95.6° D. 269.5°



30° - 16.5° + 18° = 31.5°
360° - 31.5° = 268.5°
33 x 0.5° = 16.5°
3 x 6 = 18°

Imp to revise
chat gpt

Q7 $\rightarrow$ Same que as Q6 but find for
4:44?



$$\rightarrow 15^\circ - 22^\circ - 6^\circ$$
$$= \underline{\underline{122^\circ}}$$

